Stochastic Processes
Spring 2025 April 15, 2025

**Homework #4: Exponential Distribution and Poisson Process**
Due: April 30, noon

**Problem 1**. Let $Y, X_1, \cdots, X_n$ be independent random variables with $Y \sim Exp(\mu)$ and $X_i \sim Exp(\theta)$ for each $i$. Compute the following.

(i) $E(\min\{Y, X_1, \cdots, X_n\}^{-1/2})$.

(ii) $E \max\{Y, X_1, \cdots, X_n\}$.

**Problem 2.** Suppose that customers are in line to receive service that is provided sequentially by a server; whenever a service is completed, the next person in line enters the service facility. However, each waiting customer will only wait an exponentially distributed time with rate $\theta$; if its service has not yet begun by this time then it will immediately depart the system. These exponential times, one for each waiting customer, are independent. In addition, the service times are independent exponential random variables with rate $\mu$. Suppose that Mr. Eaton is presently being served and there are $n$ other customers are waiting in line. Find the probability that every of the $n$ customers leaves before Eaton's service is completed.

**Problem 3**. Consider a post office with three clerks. Four people, A, B, C and D, enter simultaneously. A, B and $C$ go directly to the clerks, and D waits until either A, B or C leaves before he begins service. What is the probability that A and B are still in the post office after the other two have left when

(a) the service time for each clerk is exactly (nonrandom) ten minutes?

(b) the service times are exponential with mean $1/\mu$?

**Problem 4** (Chapter 5, Exercise 50). The number of hours between successive train arrivals at the station is uniformly distributed in $(0, 1)$. Passengers arrive according to a Poisson process with rate 7 per hour. Suppose a train has just left the station. Let $X$ denote the number of people who get on the next train. Find

(a) $E[X]$.

(b) $Var(X)$.

**Problem 5** (Chapter 5, Exercise 59). Cars pass an intersection according to a Poisson process with rate $\lambda$. There are 4 types of cars, and each passing car is, independently, type $i$ with probability $p_i$, $\sum_{i=1}^{4} p_i = 1$.

(a) Find the probability that at least one of each of car types 1, 2, 3 but none of type 4 have passed by time $t$.

(b). Given that exactly 6 cars of type 1 or 2 passed by time $t$, find the probability that 4 of them were type 1.

**Problem 6** (Numerical Experiment (Simulation of Poisson Process). In the lecture, we have introduced two ways to simulate a Poisson processes. In this exercise, you need to implement the two approaches via `Python` and validate that your code is correct.

In the attachment, you will find codes to generate exponential, Poisson and uniform distributions. You can use those codes to implement your simulation. Besides, it also contains codes to plot exponential and Poisson distribution functions. You can use those codes to validate your simulation results.

1. Write two functions `Poisson1` and `Poisson2` to simulate the sequence of arrival times of events on $[0, 1]$ for a Poisson process with rate $\lambda$, where $\lambda$ is the input to your functions. In `Poisson1`, the simulation is based on generating i.i.d. inter-arrival times. In `Poisson2`, you first generate the total number of arrivals and then the conditional distribution of arrival times.

2. Set $\lambda = 10$. Run `Poisson1` for 10000 rounds, record the total number of arrivals in each round. Plot the empirical distribution of the simulated number of arrivals, and validate your codes by comparing the empirical distribution with the theoretic distribution.

3. Set $\lambda = 10$. Run `Poisson2` for 10000 rounds, record the first arrival time. (What if there is no arrival on $[0, 1]$?) Plot the empirical distribution of the first arrival time, and validate your codes by comparing the empirical distribution with the theoretic distribution.